Alex Mutua

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PROFILE

I am a Computer Science and Data Science student at Williams College, with a diverse background across two continents. I have led student communities in Kenya and conducted national-level economic research at the Kenya National Bureau of Statistics. I am passionate about developing software that addresses real-world problems and have hands-on experience in full-stack development, AI-assisted tools, and data analysis. I am committed to expanding access to computing and fostering inclusive, collaborative technology communities.

EDUCATION

Williams College, Bachelor of Arts — Williamstown, Massachusetts *Graduating June 2029*
Major: Computer Science | Minor: Data Science
Relevant Coursework: Introduction to Computer Science (Object-Oriented Programming, Algorithms), Data Structures, Algorithm Design, Recursion, Data Abstraction

Moringa Institute of Technology — Kenya *September 2022*
Certificate: Software Engineering

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, R, Bash Scripting
Frameworks & Tools: React, Node.js, Git, GitHub, REDCap, Microsoft Excel
Web Technologies: HTML, CSS, REST APIs
Operating Systems: Linux, Ubuntu, macOS
Interests: Software Engineering, AI-assisted Development, Systems Design, Data Science

EXPERIENCE

Kenya National Bureau of Statistics (KNBS) — Research Assistant *June 2025 – August 2025*
Nairobi, Kenya • National Government Agency

- Supported national-level economic surveys contributing to the computation of Kenya's GDP and Consumer Price Index (CPI) through rigorous data collection and validation.
- Analyzed and validated field-generated datasets, ensuring accuracy and integrity of statistics used in official national reporting.
- Performed data entry and management using REDCap and Microsoft Excel, processing large-scale population and social sector datasets.
- Compiled population projections and generated statistics related to the social sector, contributing to evidence-based policy research.

HelloMapper — Software Engineer Intern *January 2025 – March 2025*

- Designed and implemented the frontend user interface of HelloMapper's website and application using HTML, CSS, and React.
- Integrated frontend forms with backend APIs to facilitate a complete user authentication flow, managing authentication, authorization, and user sessions.
- Tested APIs with mock data, debugged response errors, and resolved UI issues using Chrome DevTools and console logs.
- Used Git and GitHub for version control and collaborated on software design and development across the team.

PROJECTS

Automated Downloads Organizer *Personal Project • Python*

- Built a Python application using shutil and pathlib that automatically organizes a downloads folder by categorizing files into typed subfolders (Images, Documents, Videos, Archives, Executables, Songs) sorted by year of download.
- Implemented smart categorization logic: certain categories (Archives, Executables, Songs) skip year-based sorting for cleaner organization, while all others are nested by creation year.
- Used os.stat and datetime to extract file birth timestamps, enabling accurate year-based directory structuring across macOS and Linux.

Terminal-Based AI & Browser Assistant *Personal Project • Bash, Python*

- Built a Bash CLI tool (go) that launches Google or DuckDuckGo searches directly from the terminal, with flags for incognito mode (-n), search engine selection (-d), and AI mode (-a).
- Integrated the Perplexity AI API (sonar-pro model) in AI mode, enabling natural language Q&A responses inline in the terminal without switching context.
- Used Python's urllib for URL encoding and API communication, and implemented safe Bash scripting practices (set -euo pipefail) for robust error handling.

LEADERSHIP & COMMUNITY IMPACT

Founder — Coding Initiative • Williams College *February 2026*

- Founded and lead a coding initiative of 11 Williams College students focused on expanding access to computing through peer-led teaching, collaborative project building, and hackathon participation.
- Organize sessions where members build real projects together, tutor peers in programming fundamentals, and compete in hackathons as a unified team.
- Created an inclusive, structured space for students at varying skill levels to grow technically and build confidence in computing.

President — Cultural Cooking Club • Williams College *November 2025 – Present*

- Lead a multicultural organization preparing and sharing dishes from diverse countries and cultures with students, staff, and community members.
- Organize campus events fostering inclusive community building, cross-cultural dialogue, and global awareness.

Vice President — Chess Club • Williams College *October 2025 – Present*

- Support organizational planning, tournament management, and mentorship of new members to promote strategic thinking.

VOLUNTEER EXPERIENCE

Chess Tutor — Center for Learning in Action (CLiA), Williamstown Elementary School *January 2026*

- Tutored 12 elementary school students in chess fundamentals through Williams College's CLiA community engagement program over 4 weeks, contributing 16 hours of instruction.
- Fostered critical thinking, patience, and sportsmanship through structured lessons and guided play, supporting students' academic and socioemotional growth.

EARLY LEADERSHIP EXPERIENCE

School Captain — Kapsabet High School, Kenya *2019 – 2022*

- Elected School Captain in 2021 after serving as Prefect since 2019; led school-wide affairs and represented the student body to school administration.
- Previously served as Chairperson of the Chess Club, Journalism Club, and Environmental Club.

HONORS & AWARDS

Hackathon Award • Mathlete Award • Chess Champion Award • Debate Award • Student Leader Award